AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	18.8 / 6.01223
Time	2026-07-15 21:00 UTC
Location	29.6863, -95.2947
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	9 / 908	47.55 to 100; mean 84.0193	69.97 at 2026-07-15 20:55Z; dt 5 min
asos	temperature	degC	9 / 908	22 to 34.4444; mean 26.5318	28 at 2026-07-15 20:55Z; dt 5 min
asos	wind_direction	degree	9 / 1201	0 to 360; mean 106.486	0 at 2026-07-15 20:55Z; dt 5 min
asos	wind_speed	m/s	9 / 1208	0 to 10.2889; mean 2.8878	0 at 2026-07-15 20:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5881.42 to 5937.36; mean 5904.38	5897.38 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	pbl_height	m	11 / 132	21.6294 to 1860.82; mean 366.522	113.722 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	precipitable_water	mm	11 / 132	42.269 to 63.2136; mean 48.6642	50.2845 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	surface_pressure	hPa	11 / 132	1011.55 to 1019.21; mean 1015.3	1018.07 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	t_850	degC	11 / 132	16.0656 to 19.7475; mean 18.2499	17.3174 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	u_10m	m/s	11 / 132	-4.73694 to 1.38902; mean -1.5836	-1.73098 at 2026-07-15 18:00Z; dt 180 min
noaa_gfs	v_10m	m/s	11 / 132	-2.16541 to 5.42479; mean 1.4262	-1.30506 at 2026-07-15 18:00Z; dt 180 min
openaq	ozone	ppm	17 / 1147	-0.003 to 0.038; mean 0.0127	0.002 at 2026-07-15 21:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 143	14 to 74; mean 38.3217	34 at 2026-07-15 21:00Z; dt 0 min
openaq	pm25	ug/m3	79 / 4546	0 to 32.7; mean 6.0478	4.2 at 2026-07-15 21:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 359	0 to 100; mean 93.6045	100 at 2026-07-15 21:01Z; dt 1.9 min
openweather	humidity	percent	5 / 359	49 to 97; mean 86.0251	76 at 2026-07-15 21:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 359	0 to 15.12; mean 0.2914	0 at 2026-07-15 21:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 359	1016 to 1021; mean 1017.79	1018 at 2026-07-15 21:01Z; dt 1.9 min
openweather	temperature	degC	5 / 359	22.54 to 34.2; mean 26.6957	27.56 at 2026-07-15 21:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 359	0 to 355; mean 130.131	113 at 2026-07-15 21:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 359	0.34 to 5.81; mean 1.8992	0.45 at 2026-07-15 21:01Z; dt 1.9 min
purpleair	pm25	ug/m3	71 / 4861	0 to 45.7; mean 7.2635	5.4 at 2026-07-15 21:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_co_column	mol/m^2	2 / 2	0.0224462 to 0.0224492; mean 0.0224	0.0224492 at 2026-07-16 18:58Z; dt 1318.6 min
sentinel5p	s5p_co_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_hcho_column	mol/m^2	1 / 1	0.000134449 to 0.000134449; mean 0.0001	0.000134449 at 2026-07-16 19:25Z; dt 1345.1 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_no2_column	mol/m^2	1 / 1	3.22068e-05 to 3.22068e-05; mean 0	3.22068e-05 at 2026-07-16 19:25Z; dt 1345.1 min
sentinel5p	s5p_no2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_o3_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
sentinel5p	s5p_so2_column	mol/m^2	1 / 1	-2.09205e-05 to -2.09205e-05; mean -0	-2.09205e-05 at 2026-07-16 19:25Z; dt 1345.1 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-15 19:45Z; dt 74.9 min
tceq	co	ppm	3 / 172	0.1 to 0.9; mean 0.2372	0.2 at 2026-07-15 21:00Z; dt 0 min
tceq	no2	ppb	12 / 757	-0.3 to 38.4; mean 7.5758	18.8 at 2026-07-15 21:00Z; dt 0 min
tceq	so2	ppb	3 / 190	-0.6 to 1.2; mean -0.2837	-0.4 at 2026-07-15 21:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-14 06Z 94.28, 12Z 96.12, 18Z 87.53, 07-15 00Z 88.3, 06Z 87.86, 12Z 81.17, 18Z 72.97, 07-16 00Z 84.8, 06Z 93.75, 12Z 75.57, 18Z 62.47, 07-17 00Z 81.28, 06Z 88.18
asos	temperature	07-14 06Z 23.95, 12Z 23.41, 18Z 25.58, 07-15 00Z 25.43, 06Z 25.13, 12Z 25.01, 18Z 27.95, 07-16 00Z 26.59, 06Z 25.45, 12Z 29.81, 18Z 32.04, 07-17 00Z 28.1, 06Z 26.7
asos	wind_direction	07-14 06Z 136, 12Z 87.06, 18Z 108.8, 07-15 00Z 75.7, 06Z 82.73, 12Z 127.2, 18Z 77.84, 07-16 00Z 79.2, 06Z 78.14, 12Z 141.8, 18Z 152.9, 07-17 00Z 146.3, 06Z 106.9
asos	wind_speed	07-14 06Z 2.943, 12Z 2.724, 18Z 2.671, 07-15 00Z 2.705, 06Z 2.676, 12Z 2.567, 18Z 2.406, 07-16 00Z 2.109, 06Z 1.654, 12Z 3.705, 18Z 5.465, 07-17 00Z 3.435, 06Z 2.159
noaa_gfs	gh_500	07-14 12Z 5899, 18Z 5900, 07-15 00Z 5890, 06Z 5892, 12Z 5884, 18Z 5896, 07-16 00Z 5896, 06Z 5909, 12Z 5906, 18Z 5926, 07-17 00Z 5919, 06Z 5935
noaa_gfs	pbl_height	07-14 12Z 124.1, 18Z 253.7, 07-15 00Z 141.4, 06Z 153.5, 12Z 186.1, 18Z 142.3, 07-16 00Z 319, 06Z 127.9, 12Z 110.3, 18Z 1625, 07-17 00Z 860.3, 06Z 354.4
noaa_gfs	precipitable_water	07-14 12Z 58.42, 18Z 51.56, 07-15 00Z 50.38, 06Z 50.1, 12Z 47.73, 18Z 50.25, 07-16 00Z 48.02, 06Z 43.75, 12Z 44.93, 18Z 45.34, 07-17 00Z 46.17, 06Z 47.33
noaa_gfs	surface_pressure	07-14 12Z 1015, 18Z 1017, 07-15 00Z 1014, 06Z 1016, 12Z 1015, 18Z 1017, 07-16 00Z 1014, 06Z 1015, 12Z 1015, 18Z 1016, 07-17 00Z 1014, 06Z 1016
noaa_gfs	t_850	07-14 12Z 16.83, 18Z 17.35, 07-15 00Z 18.3, 06Z 17.53, 12Z 18.71, 18Z 17.37, 07-16 00Z 19.05, 06Z 19.22, 12Z 18.87, 18Z 17.82, 07-17 00Z 18.64, 06Z 19.3
noaa_gfs	u_10m	07-14 12Z -0.6575, 18Z -2.077, 07-15 00Z -1.727, 06Z -2.203, 12Z -2.582, 18Z -0.7246, 07-16 00Z -1.768, 06Z -1.734, 12Z -0.4817, 18Z -1.54, 07-17 00Z -2.477, 06Z -1.031

Source	Metric	Values
noaa_gfs	v_10m	07-14 12Z 1.35, 18Z -0.5181, 07-15 00Z 0.3779, 06Z -0.7867, 12Z 0.9323, 18Z -0.3978, 07-16 00Z 1.357, 06Z 1.628, 12Z 1.857, 18Z 4.222, 07-17 00Z 4.468, 06Z 2.626
openaq	ozone	07-14 06Z 0.007667, 12Z 0.01045, 18Z 0.01501, 07-15 00Z 0.01197, 06Z 0.007833, 12Z 0.01287, 18Z 0.01966, 07-16 00Z 0.01489, 06Z 0.003919, 12Z 0.009939, 18Z 0.0219, 07-17 00Z 0.01324, 06Z 0.009698
openaq	pm10	07-14 06Z 28.83, 12Z 19.5, 18Z 21.75, 07-15 00Z 24.25, 06Z 28.42, 12Z 48.91, 18Z 33.92, 07-16 00Z 42.75, 06Z 51.58, 12Z 56.08, 18Z 52, 07-17 00Z 43.5, 06Z 40.38
openaq	pm25	07-14 06Z 4.574, 12Z 3.461, 18Z 4.584, 07-15 00Z 4.715, 06Z 4.704, 12Z 5.603, 18Z 5.996, 07-16 00Z 6.576, 06Z 10.04, 12Z 11.41, 18Z 6.704, 07-17 00Z 6.582, 06Z 5.584
openweather	cloud_cover	07-14 06Z 98.14, 12Z 99.71, 18Z 99.87, 07-15 00Z 99.97, 06Z 100, 12Z 100, 18Z 100, 07-16 00Z 99.03, 06Z 94.77, 12Z 99.67, 18Z 98.4, 07-17 00Z 44.37, 06Z 75.5
openweather	humidity	07-14 06Z 94.21, 12Z 95.19, 18Z 92.23, 07-15 00Z 90.53, 06Z 92.24, 12Z 86.68, 18Z 79.2, 07-16 00Z 83.77, 06Z 92.97, 12Z 83.33, 18Z 66.03, 07-17 00Z 79.4, 06Z 87.64
openweather	precipitation	07-14 06Z 1.534, 12Z 1.84, 18Z 0.301, 07-15 00Z 0, 06Z 0, 12Z 0.5113, 18Z 0.041, 07-16 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-17 00Z 0, 06Z 0
openweather	pressure	07-14 06Z 1016, 12Z 1018, 18Z 1018, 07-15 00Z 1017, 06Z 1017, 12Z 1019, 18Z 1018, 07-16 00Z 1017, 06Z 1018, 12Z 1019, 18Z 1018, 07-17 00Z 1018, 06Z 1019
openweather	temperature	07-14 06Z 24.57, 12Z 23.28, 18Z 25.11, 07-15 00Z 25.41, 06Z 24.94, 12Z 25.15, 18Z 27.01, 07-16 00Z 27.07, 06Z 25.41, 12Z 29.12, 18Z 33.16, 07-17 00Z 28.9, 06Z 27.14
openweather	wind_direction	07-14 06Z 151.5, 12Z 164.6, 18Z 120.7, 07-15 00Z 71.7, 06Z 101.7, 12Z 139.1, 18Z 105.2, 07-16 00Z 124.6, 06Z 143, 12Z 154.9, 18Z 155.2, 07-17 00Z 136, 06Z 135.1
openweather	wind_speed	07-14 06Z 1.441, 12Z 2.031, 18Z 1.482, 07-15 00Z 1.853, 06Z 1.493, 12Z 1.667, 18Z 2.515, 07-16 00Z 1.716, 06Z 1.639, 12Z 1.863, 18Z 2.737, 07-17 00Z 2.168, 06Z 1.751
purpleair	pm25	07-14 06Z 5.145, 12Z 3.648, 18Z 5.262, 07-15 00Z 5.223, 06Z 5.152, 12Z 6.768, 18Z 6.637, 07-16 00Z 8.205, 06Z 12.28, 12Z 13.29, 18Z 8.049, 07-17 00Z 7.99, 06Z 7.815
sentinel5p	s5p_aer_ai_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_ch4_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_co_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_hcho_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_no2_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_o3_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	s5p_so2_granule_available	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
tceq	co	07-14 06Z 0.2111, 12Z 0.2, 18Z 0.2667, 07-15 00Z 0.17, 06Z 0.1444, 12Z 0.3, 18Z 0.2222, 07-16 00Z 0.34, 06Z 0.2611, 12Z 0.3214, 18Z 0.2083, 07-17 00Z 0.2333, 06Z 0.1625
tceq	no2	07-14 06Z 6.825, 12Z 9.866, 18Z 8.889, 07-15 00Z 7.682, 06Z 7.693, 12Z 9.772, 18Z 10.24, 07-16 00Z 10.26, 06Z 7.385, 12Z 6.061, 18Z 3.566, 07-17 00Z 3.7, 06Z 4.67
tceq	so2	07-14 06Z -0.3778, 12Z -0.3111, 18Z -0.2667, 07-15 00Z -0.2778, 06Z -0.3187, 12Z -0.2611, 18Z -0.1167, 07-16 00Z -0.1222, 06Z -0.35, 12Z -0.3222, 18Z -0.3056, 07-17 00Z -0.3, 06Z -0.3583

Per-site averages

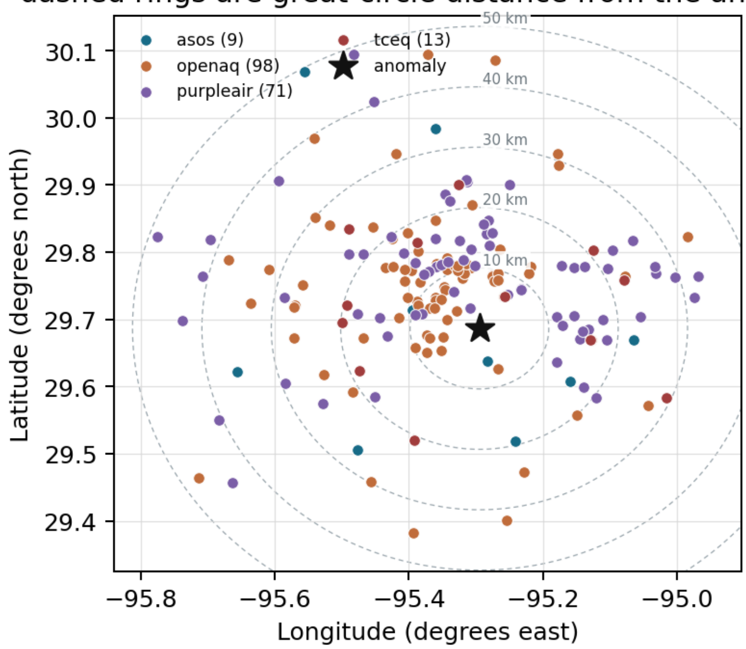
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	HOU (5.555 km) 82.22, MCJ (10.163 km) 83.02, EFD (15.811 km) 84.78, LVJ (19.307 km) 82.42, T41 (22.362 km) 83.55, AXH (26.678 km) 81.79, IAH (33.753 km) 87.24, SGR (35.675 km) 86.67, +1 more

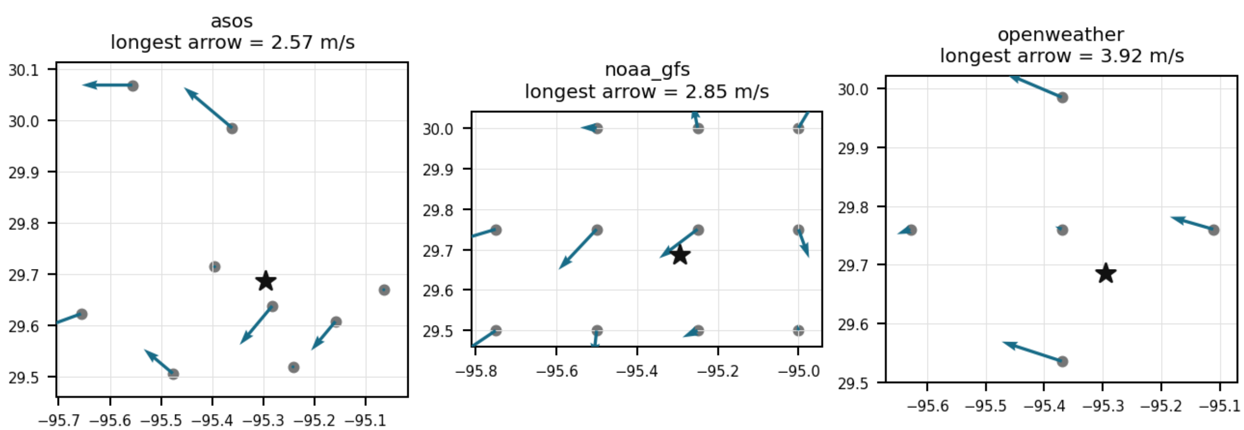
Source	Metric	Values
asos	temperature	HOU (5.555 km) 26.78, MCJ (10.163 km) 26.77, EFD (15.811 km) 26.73, LVJ (19.307 km) 26.47, T41 (22.362 km) 27.01, AXH (26.678 km) 26.36, IAH (33.753 km) 26.17, SGR (35.675 km) 26.32, +1 more
asos	wind_direction	HOU (5.555 km) 121.6, MCJ (10.163 km) 132.3, EFD (15.811 km) 88.1, LVJ (19.307 km) 92.29, T41 (22.362 km) 123.5, AXH (26.678 km) 78.46, IAH (33.753 km) 109, SGR (35.675 km) 116.2, +1 more
asos	wind_speed	HOU (5.555 km) 3.322, MCJ (10.163 km) 4.329, EFD (15.811 km) 2.825, LVJ (19.307 km) 2.326, T41 (22.362 km) 3.561, AXH (26.678 km) 1.432, IAH (33.753 km) 3.351, SGR (35.675 km) 2.958, +1 more
noaa_gfs	gh_500	gfs:29.75,-95.25 (8.297 km) 5905, gfs:29.75,-95.50 (21.05 km) 5904, gfs:29.50,-95.25 (21.162 km) 5905, gfs:29.50,-95.50 (28.689 km) 5905, gfs:29.75,-95.00 (29.33 km) 5905, gfs:30.00,-95.25 (35.148 km) 5904, gfs:29.50,-95.00 (35.231 km) 5906, gfs:30.00,-95.50 (40.109 km) 5903, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (8.297 km) 371.8, gfs:29.75,-95.50 (21.05 km) 458.6, gfs:29.50,-95.25 (21.162 km) 285.9, gfs:29.50,-95.50 (28.689 km) 322.1, gfs:29.75,-95.00 (29.33 km) 330.1, gfs:30.00,-95.25 (35.148 km) 410.9, gfs:29.50,-95.00 (35.231 km) 400.4, gfs:30.00,-95.50 (40.109 km) 483, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (8.297 km) 48.91, gfs:29.75,-95.50 (21.05 km) 48.81, gfs:29.50,-95.25 (21.162 km) 48.61, gfs:29.50,-95.50 (28.689 km) 48.92, gfs:29.75,-95.00 (29.33 km) 48.64, gfs:30.00,-95.25 (35.148 km) 48.82, gfs:29.50,-95.00 (35.231 km) 48.51, gfs:30.00,-95.50 (40.109 km) 48.43, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (8.297 km) 1016, gfs:29.75,-95.50 (21.05 km) 1015, gfs:29.50,-95.25 (21.162 km) 1016, gfs:29.50,-95.50 (28.689 km) 1015, gfs:29.75,-95.00 (29.33 km) 1017, gfs:30.00,-95.25 (35.148 km) 1015, gfs:29.50,-95.00 (35.231 km) 1017, gfs:30.00,-95.50 (40.109 km) 1013, +3 more
noaa_gfs	t_850	gfs:29.75,-95.25 (8.297 km) 18.19, gfs:29.75,-95.50 (21.05 km) 18.19, gfs:29.50,-95.25 (21.162 km) 18.44, gfs:29.50,-95.50 (28.689 km) 18.39, gfs:29.75,-95.00 (29.33 km) 18.27, gfs:30.00,-95.25 (35.148 km) 18.15, gfs:29.50,-95.00 (35.231 km) 18.39, gfs:30.00,-95.50 (40.109 km) 18.01, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (8.297 km) -1.376, gfs:29.75,-95.50 (21.05 km) -1.477, gfs:29.50,-95.25 (21.162 km) -1.802, gfs:29.50,-95.50 (28.689 km) -1.851, gfs:29.75,-95.00 (29.33 km) -1.35, gfs:30.00,-95.25 (35.148 km) -1.434, gfs:29.50,-95.00 (35.231 km) -1.717, gfs:30.00,-95.50 (40.109 km) -1.421, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (8.297 km) 1.055, gfs:29.75,-95.50 (21.05 km) 1.178, gfs:29.50,-95.25 (21.162 km) 1.652, gfs:29.50,-95.50 (28.689 km) 1.462, gfs:29.75,-95.00 (29.33 km) 1.358, gfs:30.00,-95.25 (35.148 km) 1.055, gfs:29.50,-95.00 (35.231 km) 2.409, gfs:30.00,-95.50 (40.109 km) 1.152, +3 more
openaq	ozone	325 (0.01 km) 0.003354, 3378 (6.399 km) 0.006866, 320 (7.255 km) 0.01179, 2039 (11.575 km) 0.01051, 1319333 (13.71 km) 0.01355, 2046 (15.8 km) 0.01399, 332 (16.16 km) 0.01838, 312 (18.679 km) 0.01373, +9 more
openaq	pm10	15852419 (9.465 km) 36.47, 271 (16.16 km) 40.2
openaq	pm25	14140751 (0.011 km) 6.072, 15990898 (4.432 km) 4.596, 16564662 (4.852 km) 4.766, 13418566 (5.486 km) 5.632, 8967123 (6.375 km) 8.641, 8967479 (6.375 km) 7.922, 15539682 (6.38 km) 7.21, 15472101 (6.665 km) 4.396, +71 more
openweather	cloud_cover	grid:center (10.975 km) 93.38, grid:south (18.24 km) 92.57, grid:east (19.553 km) 96.74, grid:west (33.267 km) 91.06, grid:north (33.994 km) 94.28
openweather	humidity	grid:center (10.975 km) 85.41, grid:south (18.24 km) 83.28, grid:east (19.553 km) 87.44, grid:west (33.267 km) 86.61, grid:north (33.994 km) 87.38
openweather	precipitation	grid:center (10.975 km) 0.1166, grid:south (18.24 km) 0.3574, grid:east (19.553 km) 0.2251, grid:west (33.267 km) 0.3876, grid:north (33.994 km) 0.3681
openweather	pressure	grid:center (10.975 km) 1018, grid:south (18.24 km) 1018, grid:east (19.553 km) 1018, grid:west (33.267 km) 1018, grid:north (33.994 km) 1018
openweather	temperature	grid:center (10.975 km) 26.98, grid:south (18.24 km) 26.76, grid:east (19.553 km) 26.67, grid:west (33.267 km) 26.5, grid:north (33.994 km) 26.57
openweather	wind_direction	grid:center (10.975 km) 127.8, grid:south (18.24 km) 151, grid:east (19.553 km) 141.1, grid:west (33.267 km) 144.2, grid:north (33.994 km) 86.6
openweather	wind_speed	grid:center (10.975 km) 1.908, grid:south (18.24 km) 2.141, grid:east (19.553 km) 2.362, grid:west (33.267 km) 1.383, grid:north (33.994 km) 1.701
purpleair	pm25	6752 (3.679 km) 4.759, 99797 (6.962 km) 6.45, 133994 (7.18 km) 0.07246, 293725 (8.647 km) 8.861, 97279 (8.764 km) 9.811, 166435 (8.846 km) 9.846, 293735 (9.495 km) 8.454, 222601 (10.473 km) 4.713, +63 more
tceq	co	482011035 (6.378 km) 0.1543, 482011039 (16.162 km) 0.1547, 482011052 (16.841 km) 0.3839

Source	Metric	Values
tceq	no2	482010416 (0.0 km) 7.306, 482011035 (6.378 km) 10.03, 482011039 (16.162 km) 5.498, 482011052 (16.841 km) 13.03, 482011066 (19.515 km) 8.732, 482010055 (19.781 km) 6.369, 480391004 (20.723 km) 4.094, 482010026 (20.843 km) 10.61, +4 more
tceq	so2	482011035 (6.378 km) -0.3937, 482011039 (16.162 km) 0.03387, 482010051 (18.688 km) -0.4813

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

On July 15, 2026, at 21:00 UTC, nitrogen dioxide concentrations at the TCEQ monitor in Houston spiked to 18.8 ppb, representing a severe statistical anomaly relative to expected levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

Regional photochemical accumulation is contradicted because openweather reported complete cloud cover near 21:00 UTC and openaq ozone remained very low at the anomaly location, conditions inconsistent with strong sunlight-driven ozone production as the main cause of the severe NO2 enhancement.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Meteorological trapping is supported because winds near the anomaly were very weak, with 0.0 m/s at the nearest asos station and 0.45 m/s in openweather at about 21:00 UTC, and because gfs indicated a shallow planetary boundary layer of about 114 m at 18:00 UTC, all of which would limit dilution of local NO2 emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

Regional photochemical ozone-smog buildup is an unlikely primary cause of the NO2 spike because ozone near the anomaly was extremely low at the event time despite warm conditions, indicating the NO2 enhancement was dominated by fresh local emissions and titration instead of oxidized afternoon regional air.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

The openaq ozone sensor located 0.01 km from the anomaly site measured 0.002 ppm at 2026-07-15T21:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

Near-zero surface wind speeds recorded by ASOS and OpenWeather around 21:00 UTC on July 15, 2026, support horizontal atmospheric stagnation as a key factor in accumulating NO2 detected by TCEQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

A fresh local combustion plume is supported because the anomalous tceq NO_2 concentration reached 18.8 ppb at 2026-07-15 21:00 UTC while co-located ozone was only 0.002 ppm at the same time, a combination consistent with recently emitted NO titrating ozone near a source rather than aged regional pollution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

The PM2.5 levels are elevated at 5.4 ug/m3, which is higher than the mean of 7.2635 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

There is no evidence to suggest that other air pollutants, such as CO, SO_2, or O_3, contributed to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

At 20:55 UTC on July 15, 2026, ASOS meteorological observations near the anomaly reported calm surface winds of 0.0 m/s.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The tceq NO2 monitor at 29.686, -95.295 measured 18.8 ppb at 2026-07-15T21:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

GFS model estimates of a low boundary layer height near 113.7 meters near the anomaly location support reduced vertical atmospheric mixing during the high NO2 event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

Coincident near-surface ozone concentrations fell to 0.002 ppm at 21:00 UTC on July 15, 2026, indicating ozone titration by primary nitric oxide under suppressed photolysis conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

OpenWeather reports of 100 percent cloud cover support diminished solar irradiance, which reduced photochemical photolysis of NO2 and corresponds with low surface ozone values observed by OpenAQ.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

Surface meteorological conditions during the anomaly were characterized by calm wind speeds and dense cloud cover, restricting atmospheric dispersion.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The air-quality chemistry at 2026-07-15 21:00 UTC is most consistent with a fresh local combustion plume because NO_2 was elevated while nearby ozone was only 0.002 ppm, indicating ozone titration by newly emitted NO_x rather than regional photochemical accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

High cloud cover and depressed ambient ozone levels suppressed the photochemical degradation of NO_2, facilitating its accumulation at the monitoring site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The NO_2 levels were also significantly higher than normal during the same period.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

The anomaly occurred on July 15th at around 21:00 UTC, with the nearest data point being 0.0 min away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

A fresh nearby combustion emission plume from mobile traffic, diesel equipment, rail, marine, or industrial combustion is a likely cause of the severe NO_2 spike because the monitor recorded 18.8 ppb NO_2 against an expected 6.0 ppb while co-located ozone was only 0.002 ppm, which is a classic near-source NO titration signature rather than aged regional pollution.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

Stagnant atmospheric conditions, characterized by near-zero wind speeds and a low planetary boundary layer height, restricted pollutant dispersion and trapped NO_2 near the ground.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

Meteorological conditions favored temporary trapping of local emissions because near-anomaly surface winds were extremely weak around the event and the modeled planetary boundary layer height was only about 114 m at 18:00 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

Weak dispersion and boundary-layer trapping are a likely cause of the elevated NO2 concentration because winds near the anomaly were very light at about 0.0 to 0.45 m/s and the modeled planetary boundary layer height was low at about 114 m, conditions that favor accumulation of locally emitted NOx near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

On July 15, 2026, at 21:00 UTC, TCEQ ground sensors recorded a severe NO2 elevation reaching 18.8 ppb at coordinates (29.686, -95.295), corresponding to a z-score of 6.17 above the expected level of 6.01 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

At 21:00 UTC on July 15, 2026, openaq sensors co-located at the anomaly site recorded a surface ozone level of 0.002 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

The anomaly detection record for the tceq NO2 observation at 2026-07-15T21:00:00+00:00 reports an expected value of 6.012230215827339 ppb, a z-score of 6.171617557009788, and a severity classification of severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

The TCEQ monitor at 29.686, -95.295 measured a severe NO2 anomaly of 18.8 ppb at 2026-07-15 21:00 UTC, far above the expected 6.01 ppb baseline.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The air quality anomaly in Houston, Texas on July 15, 2026 was characterized by elevated levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

Primary emissions from nearby industrial processes or urban vehicular traffic continuously supplied nitrogen oxides into the poorly mixed surface layer.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.